参考文献

全书

337 条文献 · GB/T 7714 著录 · 按正文首次引用编号

总数	篇幅	编号
337 条著录	25 页通排	10 章共用一份



参考文献

1 **RADFORD A, WU J, CHILD R, et al.** Language models are unsupervised multitask learners[Z/OL]. (2019/02/14) [2026/04/23]. https://openai.com/index/better-language-models/.

2 **BROWN T B, MANN B, RYDER N, et al.** Language Models are Few-Shot Learners[PP/OL]. arXiv. (2020/05/28) [2026/04/23]. https://arxiv.org/abs/2005.14165. arXiv:2005.14165.

3 **JIANG J, WANG F, SHEN J, et al.** A Survey on Large Language Models for Code Generation [PP/OL]. arXiv. (2024/06/01) [2026/04/23]. https://arxiv.org/abs/2406.00515. arXiv:2406.00515.

4 **AHN J, VERMA R, LOU R, et al.** Large Language Models for Mathematical Reasoning: Progresses and Challenges[PP/OL]. arXiv. (2024/01/31) [2026/04/23]. https://arxiv.org/abs/2402.00157. arXiv:2402.00157.

5 **DAM S K, HONG C S, QIAO Y, ZHANG C.** A Complete Survey on LLM-based AI Chatbots [PP/OL]. arXiv. (2024/06/17) [2026/04/23]. https://arxiv.org/abs/2406.16937. arXiv:2406.16937.

6 **QU C, DAI S, WEI X, et al.** Tool Learning with Large Language Models: A Survey[PP/OL]. arXiv. (2024/05/28) [2026/04/23]. https://arxiv.org/abs/2405.17935. arXiv:2405.17935.

7 **GAO Y, XIONG Y, GAO X, et al.** Retrieval-Augmented Generation for Large Language Models: A Survey[PP/OL]. arXiv. (2023/12/18) [2026/04/23]. https://arxiv.org/abs/2312.10997. arXiv:2312.10997.

8 **WANG L, MA C, FENG X, et al.** A Survey on Large Language Model based Autonomous Agents [PP/OL]. arXiv. (2023/08/22) [2026/04/23]. https://arxiv.org/abs/2308.11432. arXiv:2308.11432.

9 **ZHANG J, HUANG J, JIN S, LU S.** Vision-Language Models for Vision Tasks: A Survey[PP/OL]. arXiv. (2023/04/03) [2026/04/23]. https://arxiv.org/abs/2304.00685. arXiv:2304.00685.

10 **VASWANI A, SHAZEER N, PARMAR N, et al.** Attention Is All You Need[PP/OL]. arXiv. (2017/06/12) [2026/04/23]. https://arxiv.org/abs/1706.03762. arXiv:1706.03762.

11 **KAPLAN J, MCCANDLISH S, HENIGHAN T, et al.** Scaling Laws for Neural Language Models [PP/OL]. arXiv. (2020/01/23) [2026/04/23]. https://arxiv.org/abs/2001.08361. arXiv:2001.08361.

12 **WEI J, TAY Y, BOMMASANI R, et al.** Emergent Abilities of Large Language Models[PP/OL]. arXiv. (2022/06/15) [2026/04/23]. https://arxiv.org/abs/2206.07682. arXiv:2206.07682.

13 **MIKOLOV T, CHEN K, CORRADO G, DEAN J.** Efficient Estimation of Word Representations in Vector Space[PP/OL]. arXiv. (2013/01/16) [2026/04/23]. https://arxiv.org/abs/1301.3781. arXiv:1301.3781.

14 **SUTSKEVER I, VINYALS O, LE Q V.** Sequence to Sequence Learning with Neural Networks [PP/OL]. arXiv. (2014/09/10) [2026/04/23]. https://arxiv.org/abs/1409.3215. arXiv:1409.3215.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

15 **HOCHREITER S, SCHMIDHUBER J.** Long short-term memory[J]. Neural Computation, 1997, 9(8): 1735-1780.

16 **CHUNG J, GULCEHRE C, CHO K, BENGIO Y.** Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling[PP/OL]. arXiv. (2014/12/11) [2026/04/23]. https://arxiv.org/abs/1412.3555. arXiv:1412.3555.

17 **BAHDANAU D, CHO K, BENGIO Y.** Neural Machine Translation by Jointly Learning to Align and Translate[PP/OL]. arXiv. (2014/09/01) [2026/04/23]. https://arxiv.org/abs/1409.0473. arXiv:1409.0473.

18 **WU Y, SCHUSTER M, CHEN Z, et al.** Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation[PP/OL]. arXiv. (2016/09/26) [2026/04/23]. https://arxiv.org/abs/1609.08144. arXiv:1609.08144.

19 **GEHRING J, AULI M, GRANGIER D, et al.** Convolutional Sequence to Sequence Learning [PP/OL]. arXiv. (2017/05/08) [2026/04/23]. https://arxiv.org/abs/1705.03122. arXiv:1705.03122.

20 **RAFFEL C, SHAZEER N, ROBERTS A, et al.** Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer[PP/OL]. arXiv. (2019/10/23) [2026/04/23]. https://arxiv.org/abs/1910.10683. arXiv:1910.10683.

21 **DEVLIN J, CHANG M W, LEE K, TOUTANOVA K.** BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding[PP/OL]. arXiv. (2018/10/11) [2026/04/23]. https://arxiv.org/abs/1810.04805. arXiv:1810.04805.

22 **RADFORD A, NARASIMHAN K, SALIMANS T, SUTSKEVER I O.** Improving language understanding by generative pre-training[Z/OL]. (2018/06/11) [2026/04/23]. https://openai.com/index/language-unsupervised/.

23 **OpenAI, ACHIAM J, ADLER S, et al.** GPT-4 Technical Report[PP/OL]. arXiv. (2023/03/15) [2026/04/23]. https://arxiv.org/abs/2303.08774. arXiv:2303.08774.

24 **TOUVRON H, LAVRIL T, IZACARD G, et al.** LLaMA: Open and Efficient Foundation Language Models[PP/OL]. arXiv. (2023/02/27) [2026/04/23]. https://arxiv.org/abs/2302.13971. arXiv:2302.13971.

25 **TOUVRON H, MARTIN L, STONE K, et al.** Llama 2: Open Foundation and Fine-Tuned Chat Models[PP/OL]. arXiv. (2023/07/18) [2026/04/23]. https://arxiv.org/abs/2307.09288. arXiv:2307.09288.

26 **GRATTAFIORI A, DUBEY A, JAUHRI A, et al.** The Llama 3 Herd of Models[PP/OL]. arXiv. (2024/07/31) [2026/04/23]. https://arxiv.org/abs/2407.21783. arXiv:2407.21783.

27 **TEAM G, ANIL R, BORGEAUD S, et al.** Gemini: A Family of Highly Capable Multimodal Models [PP/OL]. arXiv. (2023/12/19) [2026/04/23]. https://arxiv.org/abs/2312.11805. arXiv:2312.11805.

28 **TEAM G, GEORGIEV P, LEI V I, et al.** Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context[PP/OL]. arXiv. (2024/03/08) [2026/04/23]. https://arxiv.org/abs/2403.05530. arXiv:2403.05530.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

29 **JIANG A Q, SABLAYROLLES A, MENSCH A, et al.** Mistral 7B[PP/OL]. arXiv. (2023/10/10) [2026/04/23]. https://arxiv.org/abs/2310.06825. arXiv:2310.06825.

30 **JIANG A Q, SABLAYROLLES A, ROUX A, et al.** Mixtral of Experts[PP/OL]. arXiv. (2024/01/08) [2026/04/23]. https://arxiv.org/abs/2401.04088. arXiv:2401.04088.

31 **BAI J, BAI S, CHU Y, et al.** Qwen Technical Report[PP/OL]. arXiv. (2023/09/28) [2026/04/23]. https://arxiv.org/abs/2309.16609. arXiv:2309.16609.

32 **YANG A, YANG B, HUI B, et al.** Qwen2 Technical Report[PP/OL]. arXiv. (2024/07/15) [2026/04/23]. https://arxiv.org/abs/2407.10671. arXiv:2407.10671.

33 **Qwen, YANG A, YANG B, et al.** Qwen2.5 Technical Report[PP/OL]. arXiv. (2024/12/19) [2026/04/23]. https://arxiv.org/abs/2412.15115. arXiv:2412.15115.

34 **DeepSeek-Ai, BI X, CHEN D, et al.** DeepSeek LLM: Scaling Open-Source Language Models with Longtermism[PP/OL]. arXiv. (2024/01/05) [2026/04/23]. https://arxiv.org/abs/2401.02954. arXiv:2401.02954.

35 **DeepSeek-Ai, LIU A, FENG B, et al.** DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model[PP/OL]. arXiv. (2024/05/07) [2026/04/23]. https://arxiv.org/abs/2405.04434. arXiv:2405.04434.

36 **DeepSeek-Ai, LIU A, FENG B, et al.** DeepSeek-V3 Technical Report[PP/OL]. arXiv. (2024/12/27) [2026/04/23]. https://arxiv.org/abs/2412.19437. arXiv:2412.19437.

37 **DeepSeek-Ai, GUO D, YANG D, et al.** DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning[PP/OL]. arXiv. (2025/01/22) [2026/04/23]. https://arxiv.org/abs/2501.12948. arXiv:2501.12948.

38 **LAN Z, CHEN M, GOODMAN S, et al.** ALBERT: A Lite BERT for Self-supervised Learning of Language Representations[PP/OL]. arXiv. (2019/09/26) [2026/04/23]. https://arxiv.org/abs/1909.11942. arXiv:1909.11942.

39 **LIU Y, OTT M, GOYAL N, et al.** RoBERTa: A Robustly Optimized BERT Pretraining Approach [PP/OL]. arXiv. (2019/07/26) [2026/04/23]. https://arxiv.org/abs/1907.11692. arXiv:1907.11692.

40 **CLARK K, LUONG M T, LE Q V, MANNING C D.** ELECTRA: Pre-training Text Encoders as Discriminators Rather Than Generators[PP/OL]. arXiv. (2020/03/23) [2026/04/23]. https://arxiv.org/abs/2003.10555. arXiv:2003.10555.

41 **HE P, LIU X, GAO J, CHEN W.** DeBERTa: Decoding-enhanced BERT with Disentangled Attention [PP/OL]. arXiv. (2020/06/05) [2026/04/23]. https://arxiv.org/abs/2006.03654. arXiv:2006.03654.

42 **RAE J W, BORGEAUD S, CAI T, et al.** Scaling Language Models: Methods, Analysis & Insights from Training Gopher[PP/OL]. arXiv. (2021/12/08) [2026/04/23]. https://arxiv.org/abs/2112.11446. arXiv:2112.11446.

43 **ASKELL A, BAI Y, CHEN A, et al.** A General Language Assistant as a Laboratory for Alignment [PP/OL]. arXiv. (2021/12/01) [2026/04/23]. https://arxiv.org/abs/2112.00861. arXiv:2112.00861.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

44 **THOPPILAN R, FREITAS D D, HALL J, et al.** LaMDA: Language Models for Dialog Applications [PP/OL]. arXiv. (2022/01/20) [2026/04/23]. https://arxiv.org/abs/2201.08239. arXiv:2201.08239.

45 **NAKANO R, HILTON J, BALAJI S, et al.** WebGPT: Browser-assisted question-answering with human feedback[PP/OL]. arXiv. (2021/12/17) [2026/04/23]. https://arxiv.org/abs/2112.09332. arXiv:2112.09332.

46 **CHOWDHERY A, NARANG S, DEVLIN J, et al.** PaLM: Scaling Language Modeling with Pathways [PP/OL]. arXiv. (2022/04/05) [2026/04/23]. https://arxiv.org/abs/2204.02311. arXiv:2204.02311.

47 **HOFFMANN J, BORGEAUD S, MENSCH A, et al.** Training Compute-Optimal Large Language Models[PP/OL]. arXiv. (2022/03/29) [2026/04/23]. https://arxiv.org/abs/2203.15556. arXiv:2203.15556.

48 **OUYANG L, WU J, JIANG X, et al.** Training language models to follow instructions with human feedback[PP/OL]. arXiv. (2022/03/04) [2026/04/23]. https://arxiv.org/abs/2203.02155. arXiv:2203.02155.

49 **ZHANG S, ROLLER S, GOYAL N, et al.** OPT: Open Pre-trained Transformer Language Models [PP/OL]. arXiv. (2022/05/02) [2026/04/23]. https://arxiv.org/abs/2205.01068. arXiv:2205.01068.

50 **DU Z, QIAN Y, LIU X, et al.** GLM: General Language Model Pretraining with Autoregressive Blank Infilling[PP/OL]. arXiv. (2021/03/18) [2026/04/23]. https://arxiv.org/abs/2103.10360. arXiv:2103.10360.

51 **YANG A, XIAO B, WANG B, et al.** Baichuan 2: Open Large-scale Language Models[PP/OL]. arXiv. (2023/09/19) [2026/04/23]. https://arxiv.org/abs/2309.10305. arXiv:2309.10305.

52 **TEAM G, MESNARD T, HARDIN C, et al.** Gemma: Open Models Based on Gemini Research and Technology[PP/OL]. arXiv. (2024/03/13) [2026/04/23]. https://arxiv.org/abs/2403.08295. arXiv:2403.08295.

53 **TEAM G, RIVIERE M, PATHAK S, et al.** Gemma 2: Improving Open Language Models at a Practical Size[PP/OL]. arXiv. (2024/07/31) [2026/04/23]. https://arxiv.org/abs/2408.00118. arXiv:2408.00118.

54 **TEAM G, KAMATH A, FERRET J, et al.** Gemma 3 Technical Report[PP/OL]. arXiv. (2025/03/25) [2026/04/23]. https://arxiv.org/abs/2503.19786. arXiv:2503.19786.

55 **HESTNESS J, NARANG S, ARDALANI N, et al.** Deep Learning Scaling is Predictable, Empirically [PP/OL]. arXiv. (2017/12/01) [2026/04/23]. https://arxiv.org/abs/1712.00409. arXiv:1712.00409.

56 **HENDRYCKS D, BURNS C, BASART S, et al.** Measuring Massive Multitask Language Understanding[PP/OL]. arXiv. (2020/09/07) [2026/04/23]. https://arxiv.org/abs/2009.03300. arXiv:2009.03300.

57 **SUTTON R.** The Bitter Lesson[Z/OL]. (2019/03/19) [2026/04/23]. http://www.incompleteideas.net/IncIdeas/BitterLesson.html.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

58 **SILVER D, HUBERT T, SCHRITTWIESER J, et al.** A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play[J]. Science, 2018, 362(6419): 1140-1144.

59 **SILVER D, HUANG A, MADDISON C J, et al.** Mastering the game of Go with deep neural networks and tree search[J]. Nature, 2016, 529(7587): 484-489.

60 **LOWE D G.** Distinctive image features from scale-invariant keypoints[J]. International Journal of Computer Vision, 2004, 60(2): 91-110.

61 **SUTSKEVER I, MARTENS J, HINTON G E.** Generating Text with Recurrent Neural Networks [C]//Proceedings of the 28th International Conference on Machine Learning (ICML). Washington: Omnipress, 2011: 1017-1024.

62 **LI J, LI D, XIONG C, HOI S.** BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation[PP/OL]. arXiv. (2022/01/28) [2026/04/23]. https://arxiv.org/abs/2201.12086. arXiv:2201.12086.

63 **ROMBACH R, BLATTMANN A, LORENZ D, et al.** High-Resolution Image Synthesis with Latent Diffusion Models[PP/OL]. arXiv. (2021/12/20) [2026/04/23]. https://arxiv.org/abs/2112.10752. arXiv:2112.10752.

64 **REIMERS N, GUREVYCH I.** Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks[PP/OL]. arXiv. (2019/08/27) [2026/04/23]. https://arxiv.org/abs/1908.10084. arXiv:1908.10084.

65 **DOSOVITSKIY A, BEYER L, KOLESNIKOV A, et al.** An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale[PP/OL]. arXiv. (2020/10/22) [2026/04/23]. https://arxiv.org/abs/2010.11929. arXiv:2010.11929.

66 **SHAZEER N, MIRHOSEINI A, MAZIARZ K, et al.** Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer[PP/OL]. arXiv. (2017/01/23) [2026/04/23]. https://arxiv.org/abs/1701.06538. arXiv:1701.06538.

67 **JIANG A Q, SABLAYROLLES A, ROUX A, et al.** Mixtral of Experts[PP/OL]. arXiv. (2024/01/08) [2026/04/23]. https://arxiv.org/abs/2401.04088. arXiv:2401.04088.

68 **PENG B, ALCAIDE E, ANTHONY Q, et al.** RWKV: Reinventing RNNs for the Transformer Era [PP/OL]. arXiv. (2023/05/22) [2026/04/23]. https://arxiv.org/abs/2305.13048. arXiv:2305.13048.

69 **GU A, DAO T.** Mamba: Linear-Time Sequence Modeling with Selective State Spaces[PP/OL]. arXiv. (2023/12/01) [2026/04/23]. https://arxiv.org/abs/2312.00752. arXiv:2312.00752.

70 **NIE S, ZHU F, YOU Z, et al.** Large Language Diffusion Models[PP/OL]. arXiv. (2025/02/14) [2026/04/23]. https://arxiv.org/abs/2502.09992. arXiv:2502.09992.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

71 **BAI Y, JONES A, NDOUSSE K, et al.** Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback[PP/OL]. arXiv. (2022/04/12) [2026/04/23]. https://arxiv.org/abs/2204.05862. arXiv:2204.05862.

72 **JIANG Z, LIN H, ZHONG Y, et al.** MegaScale: Scaling Large Language Model Training to More Than 10,000 GPUs[PP/OL]. arXiv. (2024/02/23) [2026/04/23]. https://arxiv.org/abs/2402.15627. arXiv:2402.15627.

73 **DAO T, FU D Y, ERMON S, et al.** FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness[PP/OL]. arXiv. (2022/05/27) [2026/04/23]. https://arxiv.org/abs/2205.14135. arXiv:2205.14135.

74 **DAO T.** FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning[PP/OL]. arXiv. (2023/07/17) [2026/04/23]. https://arxiv.org/abs/2307.08691. arXiv:2307.08691.

75 **KWON W, LI Z, ZHUANG S, et al.** Efficient Memory Management for Large Language Model Serving with PagedAttention[PP/OL]. arXiv. (2023/09/12) [2026/04/23]. https://arxiv.org/abs/2309.06180. arXiv:2309.06180.

76 **STERN M, SHAZEER N, USZKOREIT J.** Blockwise Parallel Decoding for Deep Autoregressive Models[PP/OL]. arXiv. (2018/11/07) [2026/04/23]. https://arxiv.org/abs/1811.03115. arXiv:1811.03115.

77 **CHEN C, BORGEAUD S, IRVING G, et al.** Accelerating Large Language Model Decoding with Speculative Sampling[PP/OL]. arXiv. (2023/02/02) [2026/04/23]. https://arxiv.org/abs/2302.01318. arXiv:2302.01318.

78 **LEVIATHAN Y, KALMAN M, MATIAS Y.** Fast Inference from Transformers via Speculative Decoding[PP/OL]. arXiv. (2022/11/30) [2026/04/23]. https://arxiv.org/abs/2211.17192. arXiv:2211.17192.

79 **XIA H, YANG Z, DONG Q, et al.** Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding[PP/OL]. arXiv. (2024/01/15) [2026/04/23]. https://arxiv.org/abs/2401.07851. arXiv:2401.07851.

80 **YAO S, ZHAO J, YU D, et al.** ReAct: Synergizing Reasoning and Acting in Language Models [PP/OL]. arXiv. (2022/10/06) [2026/04/23]. https://arxiv.org/abs/2210.03629. arXiv:2210.03629.

81 **WEBER M, FU D, ANTHONY Q, et al.** RedPajama: an Open Dataset for Training Large Language Models[PP/OL]. arXiv. (2024/11/19) [2026/04/23]. https://arxiv.org/abs/2411.12372. arXiv:2411.12372.

82 **PENEDO G, KYDLÍCEK H, ALLAL L B, et al.** The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale[PP/OL]. arXiv. (2024/06/25) [2026/04/23]. https://arxiv.org/abs/2406.17557. arXiv:2406.17557.

83 **LI J, FANG A, SMYRNIS G, et al.** DataComp-LM: In search of the next generation of training sets for language models[PP/OL]. arXiv. (2024/06/17) [2026/04/23]. https://arxiv.org/abs/2406.11794. arXiv:2406.11794.

84 **XU G, LIU J, YAN M, et al.** CValues: Measuring the Values of Chinese Large Language Models from Safety to Responsibility[PP/OL]. arXiv. (2023/07/19) [2026/04/23]. https://arxiv.org/abs/2307.09705. arXiv:2307.09705.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

85 **GOLDBERG Y, LEVY O.** word2vec Explained: deriving Mikolov et al.'s negative-sampling word-embedding method[PP/OL]. arXiv. (2014/02/15) [2026/04/23]. https://arxiv.org/abs/1402.3722. arXiv:1402.3722.

86 **RADFORD A, KIM J W, HALLACY C, et al.** Learning Transferable Visual Models From Natural Language Supervision[PP/OL]. arXiv. (2021/02/26) [2026/04/23]. https://arxiv.org/abs/2103.00020. arXiv:2103.00020.

87 **RAMESH A, PAVLOV M, GOH G, et al.** Zero-Shot Text-to-Image Generation[PP/OL]. arXiv. (2021/02/24) [2026/04/23]. https://arxiv.org/abs/2102.12092. arXiv:2102.12092.

88 **SU J, LU Y, PAN S, et al.** RoFormer: Enhanced Transformer with Rotary Position Embedding [PP/OL]. arXiv. (2021/04/20) [2026/04/23]. https://arxiv.org/abs/2104.09864. arXiv:2104.09864.

89 **HE K, ZHANG X, REN S, SUN J.** Deep Residual Learning for Image Recognition[PP/OL]. arXiv. (2015/12/10) [2026/04/23]. https://arxiv.org/abs/1512.03385. arXiv:1512.03385.

90 **BA J L, KIROS J R, HINTON G E.** Layer Normalization[PP/OL]. arXiv. (2016/07/21) [2026/04/23]. https://arxiv.org/abs/1607.06450. arXiv:1607.06450.

91 **KAZEMNEJAD A, PADHI I, RAMAMURTHY K N, et al.** The Impact of Positional Encoding on Length Generalization in Transformers[PP/OL]. arXiv. (2023/05/31) [2026/04/23]. https://arxiv.org/abs/2305.19466. arXiv:2305.19466.

92 **ERHAN D, COURVILLE A, BENGIO Y, VINCENT P.** Why does unsupervised pre-training help deep learning?[J]. Journal of Machine Learning Research, 2010, 11(19): 625-660.

93 **JELINEK F, MERCER R L.** Interpolated estimation of Markov source parameters from sparse data[C]//Pattern Recognition in Practice: Proceedings of an International Workshop. Amsterdam: North-Holland, 1980: 381-397.

94 **COTTRELL G W, MUNRO P, ZIPSER D.** Learning internal representation from gray-scale images: An example of extensional programming[C]//Proceedings of the Ninth Annual Conference of the Cognitive Science Society. Seattle: Lawrence Erlbaum Associates, 1987: 50-57.

95 **SCHMIDHUBER J.** Learning complex, extended sequences using the principle of history compression [J]. Neural Computation, 1992, 4(2): 234-242.

96 **BOURLARD H, KAMP Y.** Auto-association by multilayer perceptrons and singular value decomposition[J]. Biological Cybernetics, 1988, 59(4-5): 291-294.

97 **BENGIO Y, DUCHARME R, VINCENT P, JAUVIN C.** A neural probabilistic language model[J]. Journal of Machine Learning Research, 2003, 3(Feb): 1137-1155.

98 **OpenAI, AGARWAL S, AHMAD L, et al.** gpt-oss-120b & gpt-oss-20b Model Card[PP/OL]. arXiv. (2025/08/08) [2026/04/23]. https://arxiv.org/abs/2508.10925. arXiv:2508.10925.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

99 **CHEN M, TWOREK J, JUN H, et al.** Evaluating Large Language Models Trained on Code[PP/OL]. arXiv. (2021/07/07) [2026/04/23]. https://arxiv.org/abs/2107.03374. arXiv:2107.03374.

100 **XIONG R, YANG Y, HE D, et al.** On Layer Normalization in the Transformer Architecture[PP/OL]. arXiv. (2020/02/12) [2026/04/23]. https://arxiv.org/abs/2002.04745. arXiv:2002.04745.

101 **SENNRICH R, HADDOW B, BIRCH A.** Neural Machine Translation of Rare Words with Subword Units[PP/OL]. arXiv. (2015/08/31) [2026/04/23]. https://arxiv.org/abs/1508.07909. arXiv:1508.07909.

102 **HOLTZMAN A, BUYS J, DU L, et al.** The Curious Case of Neural Text Degeneration[PP/OL]. arXiv. (2019/04/22) [2026/04/23]. https://arxiv.org/abs/1904.09751. arXiv:1904.09751.

103 **KIRILLOV A, MINTUN E, RAVI N, et al.** Segment Anything[PP/OL]. arXiv. (2023/04/05) [2026/04/23]. https://arxiv.org/abs/2304.02643. arXiv:2304.02643.

104 **BAEVSKI A, ZHOU H, MOHAMED A, AULI M.** wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations[PP/OL]. arXiv. (2020/06/20) [2026/04/23]. https://arxiv.org/abs/2006.11477. arXiv:2006.11477.

105 **HUTTER M.** Universal artificial intelligence: Sequential decisions based on algorithmic probability [M]. Berlin: Springer, 2005.

106 **DELÉTANG G, RUOSS A, DUQUENNE P A, et al.** Language Modeling Is Compression[PP/OL]. arXiv. (2023/09/19) [2026/04/23]. https://arxiv.org/abs/2309.10668. arXiv:2309.10668.

107 **HUANG Y, ZHANG J, SHAN Z, HE J.** Compression Represents Intelligence Linearly[PP/OL]. arXiv. (2024/04/15) [2026/04/23]. https://arxiv.org/abs/2404.09937. arXiv:2404.09937.

108 **SHANNON C E.** A mathematical theory of communication[J]. Bell System Technical Journal, 1948, 27(3): 379-423.

109 **PASCO R C.** Source coding algorithms for fast data compression (Ph.D. Thesis abstr.)[J]. IEEE Transactions on Information Theory, 1977, 23(4): 548.

110 **RISSANEN J.** Generalized Kraft inequality and arithmetic coding[J]. IBM Journal of Research and Development, 1976, 20(3): 198-203.

111 **HOWARD P G, VITTER J S.** Analysis of arithmetic coding for data compression[C]//Data Com-pression Conference, DCC 1991. Snowbird: IEEE Computer Society, 1991: 3-12.

112 **MACKAY D J.** Information theory, inference and learning algorithms[M]. Cambridge: Cambridge university press, 2003.

113 **CUI Y, YANG Z, YAO X.** Efficient and Effective Text Encoding for Chinese LLaMA and Alpaca [PP/OL]. arXiv. (2023/04/17) [2026/04/23]. https://arxiv.org/abs/2304.08177. arXiv:2304.08177.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

114 **ROZIÈRE B, GEHRING J, GLOECKLE F, et al.** Code Llama: Open Foundation Models for Code [PP/OL]. arXiv. (2023/08/24) [2026/04/23]. https://arxiv.org/abs/2308.12950. arXiv:2308.12950.

115 **WANG Y, KORDI Y, MISHRA S, et al.** Self-Instruct: Aligning Language Models with Self-Generated Instructions[PP/OL]. arXiv. (2022/12/20) [2026/04/23]. https://arxiv.org/abs/2212.10560. arXiv:2212.10560.

116 **LIU H, LI C, WU Q, LEE Y J.** Visual Instruction Tuning[PP/OL]. arXiv. (2023/04/17) [2026/04/23]. https://arxiv.org/abs/2304.08485. arXiv:2304.08485.

117 **YANG H, LIU X Y, WANG C D.** FinGPT: Open-Source Financial Large Language Models[PP/OL]. arXiv. (2023/06/09) [2026/04/23]. https://arxiv.org/abs/2306.06031. arXiv:2306.06031.

118 **WANG H, LIU C, XI N, et al.** HuaTuo: Tuning LLaMA Model with Chinese Medical Knowledge [PP/OL]. arXiv. (2023/04/14) [2026/04/23]. https://arxiv.org/abs/2304.06975. arXiv:2304.06975.

119 **INAN H, UPASANI K, CHI J, et al.** Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations[PP/OL]. arXiv. (2023/12/07) [2026/04/23]. https://arxiv.org/abs/2312.06674. arXiv:2312.06674.

120 **RAJBHANDARI S, RASLEY J, RUWASE O, HE Y.** ZeRO: Memory Optimizations Toward Training Trillion Parameter Models[PP/OL]. arXiv. (2019/10/04) [2026/04/23]. https://arxiv.org/abs/1910.02054. arXiv:1910.02054.

121 **ZHENG Y, ZHANG R, ZHANG J, et al.** LlamaFactory: Unified Efficient Fine-Tuning of 100+ Language Models[PP/OL]. arXiv. (2024/03/20) [2026/04/23]. https://arxiv.org/abs/2403.13372. arXiv:2403.13372.

122 **GROENEVELD D, BELTAGY I, WALSH P, et al.** OLMo: Accelerating the Science of Language Models[PP/OL]. arXiv. (2024/02/01) [2026/04/23]. https://arxiv.org/abs/2402.00838. arXiv:2402.00838.

123 **OLMO T, WALSH P, SOLDAINI L, et al.** 2 OLMo 2 Furious[PP/OL]. arXiv. (2024/12/31) [2026/04/23]. https://arxiv.org/abs/2501.00656. arXiv:2501.00656.

124 **ZHANG B, SENNRICH R.** Root Mean Square Layer Normalization[PP/OL]. arXiv. (2019/10/16) [2026/04/23]. https://arxiv.org/abs/1910.07467. arXiv:1910.07467.

125 **AINSLIE J, LEE-THORP J, de JONG M, et al.** GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints[DB/OL]. arXiv. (2023/05/22) [2026/04/23]. https://arxiv.org/abs/2305.13245. arXiv: 2305.13245.

126 **SHAZEER N.** GLU Variants Improve Transformer[PP/OL]. arXiv. (2020/02/12) [2026/04/23]. https://arxiv.org/abs/2002.05202. arXiv:2002.05202.

127 **HENDRYCKS D, GIMPEL K.** Gaussian Error Linear Units (GELUs)[PP/OL]. arXiv. (2016/06/27) [2026/04/23]. https://arxiv.org/abs/1606.08415. arXiv:1606.08415.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

128 **SHAZEER N.** Fast Transformer Decoding: One Write-Head is All You Need[PP/OL]. arXiv. (2019/11/06) [2026/04/23]. https://arxiv.org/abs/1911.02150. arXiv:1911.02150.

129 **DAUPHIN Y N, FAN A, AULI M, GRANGIER D.** Language Modeling with Gated Convolutional Networks[PP/OL]. arXiv. (2016/12/23) [2026/04/23]. https://arxiv.org/abs/1612.08083. arXiv:1612.08083.

130 **YANG Y, CAO Z, CHEN Q, et al.** KVSharer: Efficient Inference via Layer-Wise Dissimilar KV Cache Sharing[PP/OL]. arXiv. (2024/10/24) [2026/04/23]. https://arxiv.org/abs/2410.18517. arXiv:2410.18517.

131 **HOOPER C, KIM S, MOHAMMADZADEH H, et al.** KVQuant: Towards 10 Million Context Length LLM Inference with KV Cache Quantization[PP/OL]. arXiv. (2024/01/31) [2026/04/23]. https://arxiv.org/abs/2401.18079. arXiv:2401.18079.

132 **ZHONG Y, LIU S, CHEN J, et al.** DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving[PP/OL]. arXiv. (2024/01/18) [2026/04/23]. https://arxiv.org/abs/2401.09670. arXiv:2401.09670.

133 **CHEN S, WONG S, CHEN L, TIAN Y.** Extending Context Window of Large Language Models via Positional Interpolation[PP/OL]. arXiv. (2023/06/27) [2026/04/23]. https://arxiv.org/abs/2306.15595. arXiv:2306.15595.

134 **CHEN Y, QIAN S, TANG H, et al.** LongLoRA: Efficient Fine-tuning of Long-Context Large Language Models[PP/OL]. arXiv. (2023/09/21) [2026/04/23]. https://arxiv.org/abs/2309.12307. arXiv:2309.12307.

135 **PENG B, QUESNELLE J, FAN H, SHIPPOLE E.** YaRN: Efficient Context Window Extension of Large Language Models[PP/OL]. arXiv. (2023/08/31) [2026/04/23]. https://arxiv.org/abs/2309.00071. arXiv:2309.00071.

136 **TAORI R O.** Alpaca: A Strong, Replicable Instruction-Following Model[Z/OL]. (2023/03/13) [2026/04/22]. https://crfm.stanford.edu/2023/03/13/alpaca.html.

137 **HOULSBY N, GIURGIU A, JASTRZEBSKI S, et al.** Parameter-Efficient Transfer Learning for NLP [PP/OL]. arXiv. (2019/02/02) [2026/04/23]. https://arxiv.org/abs/1902.00751. arXiv:1902.00751.

138 **HU E J, SHEN Y, WALLIS P, et al.** LoRA: Low-Rank Adaptation of Large Language Models [PP/OL]. arXiv. (2021/06/17) [2026/04/23]. https://arxiv.org/abs/2106.09685. arXiv:2106.09685.

139 **LI X L, LIANG P.** Prefix-Tuning: Optimizing Continuous Prompts for Generation[PP/OL]. arXiv. (2021/01/01) [2026/04/23]. https://arxiv.org/abs/2101.00190. arXiv:2101.00190.

140 **KUMAR A, RAGHUNATHAN A, JONES R, et al.** Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution[PP/OL]. arXiv. (2022/02/21) [2026/04/23]. https://arxiv.org/abs/2202.10054. arXiv:2202.10054.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

141 **QI X, ZENG Y, XIE T, et al.** Fine-tuning Aligned Language Models Compromises Safety, Even When Users Do Not Intend To![PP/OL]. arXiv. (2023/10/05) [2026/04/23]. https://arxiv.org/abs/2310.03693. arXiv:2310.03693.

142 **GHOSH S, EVURU C K R, KUMAR S, et al.** A Closer Look at the Limitations of Instruction Tuning [PP/OL]. arXiv. (2024/02/03) [2026/04/23]. https://arxiv.org/abs/2402.05119. arXiv:2402.05119.

143 **COLEMAN E N, QUARANTIELLO L, LIU Z, et al.** Parameter-Efficient Continual Fine-Tuning: A Survey[PP/OL]. arXiv. (2025/04/18) [2026/04/23]. https://arxiv.org/abs/2504.13822. arXiv:2504.13822.

144 **MAO Y, GE Y, FAN Y, et al.** A Survey on LoRA of Large Language Models[PP/OL]. arXiv. (2024/07/08) [2026/04/23]. https://arxiv.org/abs/2407.11046. arXiv:2407.11046.

145 **WANG S, YU L, LI J.** LoRA-GA: Low-Rank Adaptation with Gradient Approximation[PP/OL]. arXiv. (2024/07/06) [2026/04/23]. https://arxiv.org/abs/2407.05000. arXiv:2407.05000.

146 **DETTMERS T, PAGNONI A, HOLTZMAN A, ZETTLEMOYER L.** QLoRA: Efficient Finetuning of Quantized LLMs[PP/OL]. arXiv. (2023/05/23) [2026/04/23]. https://arxiv.org/abs/2305.14314. arXiv:2305.14314.

147 **ZHAO J, ZHANG Z, CHEN B, et al.** GaLore: Memory-Efficient LLM Training by Gradient Low-Rank Projection[PP/OL]. arXiv. (2024/03/06) [2026/04/23]. https://arxiv.org/abs/2403.03507. arXiv:2403.03507.

148 **WANG Z, LIANG J, HE R, et al.** LoRA-Pro: Are Low-Rank Adapters Properly Optimized?[PP/OL]. arXiv. (2024/07/25) [2026/04/23]. https://arxiv.org/abs/2407.18242. arXiv:2407.18242.

149 **WEI J, WANG X, SCHUURMANS D, et al.** Chain-of-Thought Prompting Elicits Reasoning in Large Language Models[PP/OL]. arXiv. (2022/01/28) [2026/04/23]. https://arxiv.org/abs/2201.11903. arXiv:2201.11903.

150 **ZHENG L, CHIANG W L, SHENG Y, et al.** Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena[PP/OL]. arXiv. (2023/06/09) [2026/04/23]. https://arxiv.org/abs/2306.05685. arXiv:2306.05685.

151 **KOJIMA T, GU S S, REID M, et al.** Large Language Models are Zero-Shot Reasoners[DB/OL]. arXiv. (2023/01/29) [2026/04/23]. https://arxiv.org/abs/2205.11916. arXiv: 2205.11916.

152 **YAO S, YU D, ZHAO J, et al.** Tree of Thoughts: Deliberate Problem Solving with Large Language Models[PP/OL]. arXiv. (2023/05/17) [2026/04/23]. https://arxiv.org/abs/2305.10601. arXiv:2305.10601.

153 **BESTA M, BLACH N, KUBICEK A, et al.** Graph of Thoughts: Solving Elaborate Problems with Large Language Models[PP/OL]. arXiv. (2023/08/18) [2026/04/23]. https://arxiv.org/abs/2308.09687. arXiv:2308.09687.

154 **COBBE K, KOSARAJU V, BAVARIAN M, et al.** Training Verifiers to Solve Math Word Problems [PP/OL]. arXiv. (2021/10/27) [2026/04/23]. https://arxiv.org/abs/2110.14168. arXiv:2110.14168.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

155 **WANG X, WEI J, SCHUURMANS D, et al.** Self-Consistency Improves Chain of Thought Reasoning in Language Models[PP/OL]. arXiv. (2022/03/21) [2026/04/23]. https://arxiv.org/abs/2203.11171. arXiv:2203.11171.

156 **VATSAL S, DUBEY H.** A Survey of Prompt Engineering Methods in Large Language Models for Different NLP Tasks[PP/OL]. arXiv. (2024/07/17) [2026/04/23]. https://arxiv.org/abs/2407.12994. arXiv:2407.12994.

157 **QU Y, ZHANG T, GARG N, KUMAR A.** Recursive Introspection: Teaching Language Model Agents How to Self-Improve[PP/OL]. arXiv. (2024/07/25) [2026/04/23]. https://arxiv.org/abs/2407.18219. arXiv:2407.18219.

158 **MADAAN A, TANDON N, GUPTA P, et al.** Self-Refine: Iterative Refinement with Self-Feedback [PP/OL]. arXiv. (2023/05/30) [2026/04/23]. https://arxiv.org/abs/2303.17651. arXiv:2303.17651.

159 **REN J, ZHAO Y, VU T, et al.** Self-Evaluation Improves Selective Generation in Large Language Models[PP/OL]. arXiv. (2023/12/14) [2026/04/23]. https://arxiv.org/abs/2312.09300. arXiv:2312.09300.

160 **ASAI A, WU Z, WANG Y, et al.** Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection[PP/OL]. arXiv. (2023/10/17) [2026/04/23]. https://arxiv.org/abs/2310.11511. arXiv:2310.11511.

161 **WANG L, YANG N, HUANG X, et al.** Text Embeddings by Weakly-Supervised Contrastive Pre-training[PP/OL]. arXiv. (2022/12/07) [2026/04/23]. https://arxiv.org/abs/2212.03533. arXiv:2212.03533.

162 **LI Z, ZHANG X, ZHANG Y, et al.** Towards General Text Embeddings with Multi-stage Contrastive Learning[PP/OL]. arXiv. (2023/08/07) [2026/04/23]. https://arxiv.org/abs/2308.03281. arXiv:2308.03281.

163 **BEHNAMGHADER P, ADLAKHA V, MOSBACH M, et al.** LLM2Vec: Large Language Models Are Secretly Powerful Text Encoders[PP/OL]. arXiv. (2024/04/09) [2026/04/23]. https://arxiv.org/abs/2404.05961. arXiv:2404.05961.

164 **LEE C, ROY R, XU M, et al.** NV-Embed: Improved Techniques for Training LLMs as Generalist Embedding Models[PP/OL]. arXiv. (2024/05/27) [2026/04/23]. https://arxiv.org/abs/2405.17428. arXiv:2405.17428.

165 **MUENNIGHOFF N, TAZI N, MAGNE L, REIMERS N.** MTEB: Massive Text Embedding Benchmark[PP/OL]. arXiv. (2022/10/13) [2026/04/23]. https://arxiv.org/abs/2210.07316. arXiv:2210.07316.

166 **SHEN Y, SONG K, TAN X, et al.** HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face[PP/OL]. arXiv. (2023/03/30) [2026/04/23]. https://arxiv.org/abs/2303.17580. arXiv:2303.17580.

167 **YANG Z, QI P, ZHANG S, et al.** HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering[PP/OL]. arXiv. (2018/09/25) [2026/04/23]. https://arxiv.org/abs/1809.09600. arXiv:1809.09600.

168 **FORD JR L R.** Solution of a ranking problem from binary comparisons[J]. The American Mathematical Monthly, 1957, 64(8): 28-33.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

169 **GU J, JIANG X, SHI Z, et al.** A Survey on LLM-as-a-Judge[PP/OL]. arXiv. (2024/11/23) [2026/04/23]. https://arxiv.org/abs/2411.15594. arXiv:2411.15594.

170 **LI Y, GUERIN F, LIN C.** An Open Source Data Contamination Report for Large Language Models [PP/OL]. arXiv. (2023/10/26) [2026/04/23]. https://arxiv.org/abs/2310.17589. arXiv:2310.17589.

171 **CHANG Y, WANG X, WANG J, et al.** A Survey on Evaluation of Large Language Models[PP/OL]. arXiv. (2023/07/06) [2026/04/23]. https://arxiv.org/abs/2307.03109. arXiv:2307.03109.

172 **CHIANG W L, ZHENG L, SHENG Y, et al.** Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference[PP/OL]. arXiv. (2024/03/07) [2026/04/23]. https://arxiv.org/abs/2403.04132. arXiv:2403.04132.

173 **HENDRYCKS D, BURNS C, KADAVATH S, et al.** Measuring Mathematical Problem Solving With the MATH Dataset[PP/OL]. arXiv. (2021/03/05) [2026/04/23]. https://arxiv.org/abs/2103.03874. arXiv:2103.03874.

174 **XIE S M, RAGHUNATHAN A, LIANG P, MA T.** An Explanation of In-context Learning as Implicit Bayesian Inference[PP/OL]. arXiv. (2021/11/03) [2026/04/23]. https://arxiv.org/abs/2111.02080. arXiv:2111.02080.

175 **WIES N, LEVINE Y, SHASHUA A.** The Learnability of In-Context Learning[PP/OL]. arXiv. (2023/03/14) [2026/04/23]. https://arxiv.org/abs/2303.07895. arXiv:2303.07895.

176 **DAI D, SUN Y, DONG L, et al.** Why Can GPT Learn In-Context? Language Models Implicitly Perform Gradient Descent as Meta-Optimizers[PP/OL]. arXiv. (2022/12/20) [2026/04/23]. https://arxiv.org/abs/2212.10559. arXiv:2212.10559.

177 **HAHN M, GOYAL N.** A Theory of Emergent In-Context Learning as Implicit Structure Induction [PP/OL]. arXiv. (2023/03/14) [2026/04/23]. https://arxiv.org/abs/2303.07971. arXiv:2303.07971.

178 **SUTTON R S.** Learning to predict by the methods of temporal differences[J]. Machine Learning, 1988, 3(1): 9-44.

179 **SUTTON R S, BARTO A G.** Reinforcement Learning: An Introduction (2nd Edition)[M]. Massachusetts: MIT Press, 2018.

180 **WATKINS C J, DAYAN P.** Q-learning[J]. Machine Learning, 1992, 8(3): 279-292.

181 **MNIH V, KAVUKCUOGLU K, SILVER D, et al.** Human-level control through deep reinforcement learning[J]. Nature, 2015, 518(7540): 529-533.

182 **LIN L J.** Self-improving reactive agents based on reinforcement learning, planning and teaching[J]. Machine Learning, 1992, 8(3): 293-321.

183 **WILLIAMS R J.** Simple statistical gradient-following algorithms for connectionist reinforcement learning[J]. Machine Learning, 1992, 8(3-4): 229-256.

参考文献

184 **SUTTON R S, MCALLESTER D, SINGH S, MANSOUR Y.** Policy gradient methods for rein-forcement learning with function approximation[C]//Advances in Neural Information Processing Systems. Denver: The MIT Press, 1999: 1057-1063.

185 **BARTO A G, SUTTON R S, ANDERSON C W.** Neuronlike adaptive elements that can solve difficult learning control problems[J]. IEEE Transactions on Systems, Man, and Cybernetics, 1983, SMC-13(5): 834-846.

186 **SCHULMAN J, LEVINE S, MORITZ P, et al.** Trust Region Policy Optimization[PP/OL]. arXiv. (2015/02/19) [2026/04/23]. https://arxiv.org/abs/1502.05477. arXiv:1502.05477.

187 **SCHULMAN J, MORITZ P, LEVINE S, et al.** High-Dimensional Continuous Control Using Generalized Advantage Estimation[PP/OL]. arXiv. (2015/06/08) [2026/04/23]. https://arxiv.org/abs/1506.02438. arXiv:1506.02438.

188 **SCHULMAN J, WOLSKI F, DHARIWAL P, et al.** Proximal Policy Optimization Algorithms [PP/OL]. arXiv. (2017/07/20) [2026/04/23]. https://arxiv.org/abs/1707.06347. arXiv:1707.06347.

189 **WEI J, BOSMA M, ZHAO V Y, et al.** Finetuned Language Models Are Zero-Shot Learners[PP/OL]. arXiv. (2021/09/03) [2026/04/23]. https://arxiv.org/abs/2109.01652. arXiv:2109.01652.

190 **CHU T, ZHAI Y, YANG J, et al.** SFT Memorizes, RL Generalizes: A Comparative Study of Foundation Model Post-training[PP/OL]. arXiv. (2025/01/28) [2026/04/23]. https://arxiv.org/abs/2501.17161. arXiv:2501.17161.

191 **JI J, QIU T, CHEN B, et al.** AI Alignment: A Comprehensive Survey[PP/OL]. arXiv. (2023/10/30) [2026/04/23]. https://arxiv.org/abs/2310.19852. arXiv:2310.19852.

192 **SHAO Z, WANG P, ZHU Q, et al.** DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models[PP/OL]. arXiv. (2024/02/05) [2026/04/23]. https://arxiv.org/abs/2402.03300. arXiv:2402.03300.

193 **LIGHTMAN H, KOSARAJU V, BURDA Y, et al.** Let's Verify Step by Step[PP/OL]. arXiv. (2023/05/31) [2026/04/23]. https://arxiv.org/abs/2305.20050. arXiv:2305.20050.

194 **WANG P, LI L, SHAO Z, et al.** Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations[PP/OL]. arXiv. (2023/12/14) [2026/04/23]. https://arxiv.org/abs/2312.08935. arXiv:2312.08935.

195 **FENG X, WAN Z, WEN M, et al.** Alphazero-like Tree-Search can Guide Large Language Model Decoding and Training[PP/OL]. arXiv. (2023/09/29) [2026/04/23]. https://arxiv.org/abs/2309.17179. arXiv:2309.17179.

196 **SNELL C, LEE J, XU K, KUMAR A.** Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters[PP/OL]. arXiv. (2024/08/06) [2026/04/23]. https://arxiv.org/abs/2408.03314. arXiv:2408.03314.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

197 **BURNS C, IZMAILOV P, KIRCHNER J H, et al.** Weak-to-Strong Generalization: Eliciting Strong Capabilities With Weak Supervision[PP/OL]. arXiv. (2023/12/14) [2026/04/23]. https://arxiv.org/abs/2312.09390. arXiv:2312.09390.

198 **XI Z, CHEN W, GUO X, et al.** The Rise and Potential of Large Language Model Based Agents: A Survey[PP/OL]. arXiv. (2023/09/14) [2026/04/23]. https://arxiv.org/abs/2309.07864. arXiv:2309.07864.

199 **DEGRAVE J, FELICI F, BUCHLI J, et al.** Magnetic control of tokamak plasmas through deep reinforcement learning[J]. Nature, 2022, 602(7897): 414-419.

200 **TURNER A M, SMITH L, SHAH R, et al.** Optimal Policies Tend to Seek Power[PP/OL]. arXiv. (2019/12/03) [2026/04/23]. https://arxiv.org/abs/1912.01683. arXiv:1912.01683.

201 **PEREZ E, RINGER S, LUKOSIŪTE K, et al.** Discovering Language Model Behaviors with Model-Written Evaluations[PP/OL]. arXiv. (2022/12/19) [2026/04/23]. https://arxiv.org/abs/2212.09251. arXiv:2212.09251.

202 **CARROLL M, CHAN A, ASHTON H, KRUEGER D.** Characterizing Manipulation from AI Systems [PP/OL]. arXiv. (2023/03/16) [2026/04/23]. https://arxiv.org/abs/2303.09387. arXiv:2303.09387.

203 **SHARMA M, TONG M, KORBAK T, et al.** Towards Understanding Sycophancy in Language Models[PP/OL]. arXiv. (2023/10/20) [2026/04/23]. https://arxiv.org/abs/2310.13548. arXiv:2310.13548.

204 **PARK P S, GOLDSTEIN S, O'GARA A, et al.** AI Deception: A Survey of Examples, Risks, and Potential Solutions[PP/OL]. arXiv. (2023/08/28) [2026/04/23]. https://arxiv.org/abs/2308.14752. arXiv:2308.14752.

205 **LEIKE J, KRUEGER D, EVERITT T, et al.** Scalable agent alignment via reward modeling: a research direction[PP/OL]. arXiv. (2018/11/19) [2026/04/23]. https://arxiv.org/abs/1811.07871. arXiv:1811.07871.

206 **PAN A, BHATIA K, STEINHARDT J.** The Effects of Reward Misspecification: Mapping and Mitigating Misaligned Models[PP/OL]. arXiv. (2022/01/10) [2026/04/23]. https://arxiv.org/abs/2201.03544. arXiv:2201.03544.

207 **SHAH R, VARMA V, KUMAR R, et al.** Goal Misgeneralization: Why Correct Specifications Aren't Enough For Correct Goals[PP/OL]. arXiv. (2022/10/04) [2026/04/23]. https://arxiv.org/abs/2210.01790. arXiv:2210.01790.

208 **BERGLUND L, STICKLAND A C, BALESNI M, et al.** Taken out of context: On measuring situational awareness in LLMs[PP/OL]. arXiv. (2023/09/01) [2026/04/23]. https://arxiv.org/abs/2309.00667. arXiv:2309.00667.

209 **NGO R, CHAN L, MINDERMANN S.** The Alignment Problem from a Deep Learning Perspective [PP/OL]. arXiv. (2022/08/30) [2026/04/23]. https://arxiv.org/abs/2209.00626. arXiv:2209.00626.

210 **HUBINGER E, van MERWIJK C, MIKULIK V, et al.** Risks from Learned Optimization in Advanced Machine Learning Systems[PP/OL]. arXiv. (2019/06/05) [2026/04/23]. https://arxiv.org/abs/1906.01820. arXiv:1906.01820.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

211 **SHEVLANE T, FARQUHAR S, GARFINKEL B, et al.** Model evaluation for extreme risks[PP/OL]. arXiv. (2023/05/24) [2026/04/23]. https://arxiv.org/abs/2305.15324. arXiv:2305.15324.

212 **DIETTERICH T G.** Steps toward robust artificial intelligence[J]. AI Magazine, 2017, 38(3): 3-24.

213 **TALEB N N.** The black swan: The impact of the highly improbable[M]. New York: Random house, 2007.

214 **HENDRYCKS D, CARLINI N, SCHULMAN J, STEINHARDT J.** Unsolved Problems in ML Safety [PP/OL]. arXiv. (2021/09/28) [2026/04/23]. https://arxiv.org/abs/2109.13916. arXiv:2109.13916.

215 **SONG Y, SHU R, KUSHMAN N, ERMON S.** Constructing Unrestricted Adversarial Examples with Generative Models[PP/OL]. arXiv. (2018/05/21) [2026/04/23]. https://arxiv.org/abs/1805.07894. arXiv:1805.07894.

216 **CHAKRABORTY A, ALAM M, DEY V, et al.** Adversarial Attacks and Defences: A Survey[PP/OL]. arXiv. (2018/09/28) [2026/04/23]. https://arxiv.org/abs/1810.00069. arXiv:1810.00069.

217 **CARLINI N, NASR M, CHOQUETTE-CHOO C A, et al.** Are aligned neural networks adversarially aligned?[PP/OL]. arXiv. (2023/06/26) [2026/04/23]. https://arxiv.org/abs/2306.15447. arXiv:2306.15447.

218 **ZOU A, WANG Z, CARLINI N, et al.** Universal and Transferable Adversarial Attacks on Aligned Language Models[PP/OL]. arXiv. (2023/07/27) [2026/04/23]. https://arxiv.org/abs/2307.15043. arXiv:2307.15043.

219 **SHAH R, FEUILLADE–MONTIXI Q, POUR S, et al.** Scalable and Transferable Black-Box Jailbreaks for Language Models via Persona Modulation[PP/OL]. arXiv. (2023/11/06) [2026/04/23]. https://arxiv.org/abs/2311.03348. arXiv:2311.03348.

220 **RÄUKER T, HO A, CASPER S, HADFIELD-MENELL D.** Toward Transparent AI: A Survey on Interpreting the Inner Structures of Deep Neural Networks[PP/OL]. arXiv. (2022/07/27) [2026/04/23]. https://arxiv.org/abs/2207.13243. arXiv:2207.13243.

221 **TURPIN M, MICHAEL J, PEREZ E, BOWMAN S R.** Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting[PP/OL]. arXiv. (2023/05/07) [2026/04/23]. https://arxiv.org/abs/2305.04388. arXiv:2305.04388.

222 **CARRANZA A, PAI D, SCHAEFFER R, et al.** Deceptive Alignment Monitoring[PP/OL]. arXiv. (2023/07/20) [2026/04/23]. https://arxiv.org/abs/2307.10569. arXiv:2307.10569.

223 **PACCHIARDI L, CHAN A J, MINDERMANN S, et al.** How to Catch an AI Liar: Lie Detection in Black-Box LLMs by Asking Unrelated Questions[PP/OL]. arXiv. (2023/09/26) [2026/04/23]. https://arxiv.org/abs/2309.15840. arXiv:2309.15840.

224 **RADHAKRISHNAN A, NGUYEN K, CHEN A, et al.** Question Decomposition Improves the Faithfulness of Model-Generated Reasoning[PP/OL]. arXiv. (2023/07/17) [2026/04/23]. https://arxiv.org/abs/2307.11768. arXiv:2307.11768.

参考文献

225 **ELHAGE N, NANDA N, OLSSON C, et al.** A Mathematical Framework for Transformer Circuits [Z/OL]. (2021/12/22) [2026/04/23]. https://transformer-circuits.pub/2021/framework/index.html.

226 **MENG K, BAU D, ANDONIAN A, BELINKOV Y.** Locating and Editing Factual Associations in GPT[PP/OL]. arXiv. (2022/02/10) [2026/04/23]. https://arxiv.org/abs/2202.05262. arXiv:2202.05262.

227 **HADFIELD-MENELL D, DRAGAN A, ABBEEL P, RUSSELL S.** The Off-Switch Game[PP/OL]. arXiv. (2016/11/24) [2026/04/23]. https://arxiv.org/abs/1611.08219. arXiv:1611.08219.

228 **CRITCH A, KRUEGER D.** AI Research Considerations for Human Existential Safety (ARCHES) [PP/OL]. arXiv. (2020/05/30) [2026/04/23]. https://arxiv.org/abs/2006.04948. arXiv:2006.04948.

229 **BOWMAN S R, HYUN J, PEREZ E, et al.** Measuring Progress on Scalable Oversight for Large Language Models[PP/OL]. arXiv. (2022/11/04) [2026/04/23]. https://arxiv.org/abs/2211.03540. arXiv:2211.03540.

230 **ZHANG B H, LEMOINE B, MITCHELL M.** Mitigating Unwanted Biases with Adversarial Learning [PP/OL]. arXiv. (2018/01/22) [2026/04/23]. https://arxiv.org/abs/1801.07593. arXiv:1801.07593.

231 **RAJI I D, GEBRU T, MITCHELL M, et al.** Saving Face: Investigating the Ethical Concerns of Facial Recognition Auditing[PP/OL]. arXiv. (2020/01/03) [2026/04/23]. https://arxiv.org/abs/2001.00964. arXiv:2001.00964.

232 **BERK R A, HEIDARI H, JABBARI S, et al.** Fairness in Criminal Justice Risk Assessments: The State of the Art[PP/OL]. arXiv. (2017/03/27) [2026/04/23]. https://arxiv.org/abs/1703.09207. arXiv:1703.09207.

233 **HENDRYCKS D, BURNS C, BASART S, et al.** Aligning AI With Shared Human Values[PP/OL]. arXiv. (2020/08/05) [2026/04/23]. https://arxiv.org/abs/2008.02275. arXiv:2008.02275.

234 **HENDRYCKS D, MAZEIKA M, ZOU A, et al.** What Would Jiminy Cricket Do? Towards Agents That Behave Morally[PP/OL]. arXiv. (2021/10/25) [2026/04/23]. https://arxiv.org/abs/2110.13136. arXiv:2110.13136.

235 **PAN A, CHAN J S, ZOU A, et al.** Do the Rewards Justify the Means? Measuring Trade-Offs Between Rewards and Ethical Behavior in the MACHIAVELLI Benchmark[PP/OL]. arXiv. (2023/04/06) [2026/04/23]. https://arxiv.org/abs/2304.03279. arXiv:2304.03279.

236 **HAGENDORFF T.** The ethics of AI ethics: An evaluation of guidelines[J]. Minds and Machines, 2020, 30(1): 99-120.

237 **WINFIELD A F, MICHAEL K, PITT J, EVERS V.** Machine ethics: The design and governance of ethical AI and autonomous systems [scanning the issue][J]. Proceedings of the IEEE, 2019, 107(3): 509-517.

238 **CHRISTIANO P, LEIKE J, BROWN T B, et al.** Deep reinforcement learning from human preferences[PP/OL]. arXiv. (2017/06/12) [2026/04/23]. https://arxiv.org/abs/1706.03741. arXiv:1706.03741.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

239 **PANDEY R, PUROHIT H, CASTILLO C, SHALIN V L.** Modeling and mitigating human annotation errors to design efficient stream processing systems with human-in-the-loop machine learning [PP/OL]. arXiv. (2020/07/07) [2026/04/23]. https://arxiv.org/abs/2007.03177. arXiv:2007.03177.

240 **CASPER S, DAVIES X, SHI C, et al.** Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback[PP/OL]. arXiv. (2023/07/27) [2026/04/23]. https://arxiv.org/abs/2307.15217. arXiv:2307.15217.

241 **TIEN J, HE J Z Y, ERICKSON Z, et al.** Causal Confusion and Reward Misidentification in Preference-Based Reward Learning[PP/OL]. arXiv. (2022/04/13) [2026/04/23]. https://arxiv.org/abs/2204.06601. arXiv:2204.06601.

242 **ANDERSON M, ANDERSON S L.** Machine ethics[M]. Cambridge: Cambridge University Press, 2011.

243 **TOLMEIJER S, KNEER M, SARASUA C, et al.** Implementations in Machine Ethics: A Survey [PP/OL]. arXiv. (2020/01/21) [2026/04/23]. https://arxiv.org/abs/2001.07573. arXiv:2001.07573.

244 **SANTURKAR S, DURMUS E, LADHAK F, et al.** Whose Opinions Do Language Models Reflect? [PP/OL]. arXiv. (2023/03/30) [2026/04/23]. https://arxiv.org/abs/2303.17548. arXiv:2303.17548.

245 **KRUEGER D, MAHARAJ T, LEIKE J.** Hidden Incentives for Auto-Induced Distributional Shift [PP/OL]. arXiv. (2020/09/19) [2026/04/23]. https://arxiv.org/abs/2009.09153. arXiv:2009.09153.

246 **THULASIDASAN S, THAPA S, DHAUBHADEL S, et al.** An Effective Baseline for Robustness to Distributional Shift[PP/OL]. arXiv. (2021/05/15) [2026/04/23]. https://arxiv.org/abs/2105.07107. arXiv:2105.07107.

247 **HENDRYCKS D, BASART S, MU N, et al.** The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization[PP/OL]. arXiv. (2020/06/29) [2026/04/23]. https://arxiv.org/abs/2006.16241. arXiv:2006.16241.

248 **LANGOSCO L, KOCH J, SHARKEY L, et al.** Goal Misgeneralization in Deep Reinforcement Learning[PP/OL]. arXiv. (2021/05/28) [2026/04/23]. https://arxiv.org/abs/2105.14111. arXiv:2105.14111.

249 **PERDOMO J C, ZRNIC T, MENDLER-DÜNNER C, HARDT M.** Performative Prediction [PP/OL]. arXiv. (2020/02/16) [2026/04/23]. https://arxiv.org/abs/2002.06673. arXiv:2002.06673.

250 **KALIMERIS D, BHAGAT S, KALYANARAMAN S, WEINSBERG U.** Preference amplification in recommender systems[C]//KDD '21: The 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. Singapore: Association for Computing Machinery (ACM), 2021: 805-815.

251 **ADOMAVICIUS G, BOCKSTEDT J, CURLEY S, ZHANG J.** Recommender systems, ground truth, and preference pollution[J]. AI Magazine, 2022, 43(2): 177-189.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

252 **KRUEGER D, CABALLERO E, JACOBSEN J H, et al.** Out-of-Distribution Generalization via Risk Extrapolation (REx)[PP/OL]. arXiv. (2020/03/02) [2026/04/23]. https://arxiv.org/abs/2003.00688. arXiv:2003.00688.

253 **LUBANA E S, BIGELOW E J, DICK R P, et al.** Mechanistic Mode Connectivity[PP/OL]. arXiv. (2022/11/15) [2026/04/23]. https://arxiv.org/abs/2211.08422. arXiv:2211.08422.

254 **BAI T, LUO J, ZHAO J, et al.** Recent Advances in Adversarial Training for Adversarial Robustness [PP/OL]. arXiv. (2021/02/02) [2026/04/23]. https://arxiv.org/abs/2102.01356. arXiv:2102.01356.

255 **DAFOE A, HUGHES E, BACHRACH Y, et al.** Open Problems in Cooperative AI[PP/OL]. arXiv. (2020/12/15) [2026/04/23]. https://arxiv.org/abs/2012.08630. arXiv:2012.08630.

256 **DAFOE A, BACHRACH Y, HADFIELD G, et al.** Cooperative AI: machines must learn to find common ground[J]. Nature, 2021, 593(7857): 33-36.

257 **ANDERLJUNG M, BARNHART J, KORINEK A, et al.** Frontier AI Regulation: Managing Emerging Risks to Public Safety[PP/OL]. arXiv. (2023/07/06) [2026/04/23]. https://arxiv.org/abs/2307.03718. arXiv:2307.03718.

258 **OLAH C, SATYANARAYAN A, JOHNSON I, et al.** The building blocks of interpretability[J]. Distill, 2018, 3(3): e10.

259 **PEREZ E, HUANG S, SONG F, et al.** Red Teaming Language Models with Language Models [PP/OL]. arXiv. (2022/02/07) [2026/04/23]. https://arxiv.org/abs/2202.03286. arXiv:2202.03286.

260 **KOESSLER L, SCHUETT J.** Risk assessment at AGI companies: A review of popular risk assessment techniques from other safety-critical industries[PP/OL]. arXiv. (2023/07/17) [2026/04/23]. https://arxiv.org/abs/2307.08823. arXiv:2307.08823.

261 **BURNS C, YE H, KLEIN D, STEINHARDT J.** Discovering Latent Knowledge in Language Models Without Supervision[PP/OL]. arXiv. (2022/12/07) [2026/04/23]. https://arxiv.org/abs/2212.03827. arXiv:2212.03827.

262 **SCHUETT J, DREKSLER N, ANDERLJUNG M, et al.** Towards best practices in AGI safety and governance: A survey of expert opinion[PP/OL]. arXiv. (2023/05/11) [2026/04/23]. https://arxiv.org/abs/2305.07153. arXiv:2305.07153.

263 **SEGER E, DREKSLER N, MOULANGE R, et al.** Open-Sourcing Highly Capable Foundation Models: An evaluation of risks, benefits, and alternative methods for pursuing open-source objectives [PP/OL]. arXiv. (2023/09/29) [2026/04/23]. https://arxiv.org/abs/2311.09227. arXiv:2311.09227.

264 **HO L, BARNHART J, TRAGER R, et al.** International Institutions for Advanced AI[PP/OL]. arXiv. (2023/07/10) [2026/04/23]. https://arxiv.org/abs/2307.04699. arXiv:2307.04699.

265 **STIENNON N, OUYANG L, WU J, et al.** Learning to summarize from human feedback[PP/OL]. arXiv. (2020/09/02) [2026/04/23]. https://arxiv.org/abs/2009.01325. arXiv:2009.01325.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

266 **ZIEGLER D M, STIENNON N, WU J, et al.** Fine-Tuning Language Models from Human Preferences [PP/OL]. arXiv. (2019/09/18) [2026/04/23]. https://arxiv.org/abs/1909.08593. arXiv:1909.08593.

267 **IBARZ B, LEIKE J, POHLEN T, et al.** Reward learning from human preferences and demonstrations in Atari[PP/OL]. arXiv. (2018/11/15) [2026/04/23]. https://arxiv.org/abs/1811.06521. arXiv:1811.06521.

268 **BÖHM F, GAO Y, MEYER C M, et al.** Better Rewards Yield Better Summaries: Learning to Summarise Without References[PP/OL]. arXiv. (2019/09/03) [2026/04/23]. https://arxiv.org/abs/1909.01214. arXiv:1909.01214.

269 **BAHDANAU D, HILL F, LEIKE J, et al.** Learning to Understand Goal Specifications by Modelling Reward[PP/OL]. arXiv. (2018/06/05) [2026/04/23]. https://arxiv.org/abs/1806.01946. arXiv:1806.01946.

270 **BLACK K, JANNER M, DU Y, et al.** Training Diffusion Models with Reinforcement Learning [PP/OL]. arXiv. (2023/05/22) [2026/04/23]. https://arxiv.org/abs/2305.13301. arXiv:2305.13301.

271 **HU J, WU X, SHEN W, et al.** OpenRLHF: An Easy-to-use, Scalable and High-performance RLHF Framework[PP/OL]. arXiv. (2024/05/20) [2026/04/23]. https://arxiv.org/abs/2405.11143. arXiv:2405.11143.

272 **RAFAILOV R, SHARMA A, MITCHELL E, et al.** Direct Preference Optimization: Your Language Model is Secretly a Reward Model[PP/OL]. arXiv. (2023/05/29) [2026/04/23]. https://arxiv.org/abs/2305.18290. arXiv:2305.18290.

273 **LI Z, XU T, ZHANG Y, et al.** ReMax: A Simple, Effective, and Efficient Reinforcement Learning Method for Aligning Large Language Models[PP/OL]. arXiv. (2023/10/16) [2026/04/23]. https://arxiv.org/abs/2310.10505. arXiv:2310.10505.

274 **AHMADIAN A, CREMER C, GALLÉ M, et al.** Back to Basics: Revisiting REINFORCE Style Optimization for Learning from Human Feedback in LLMs[PP/OL]. arXiv. (2024/02/22) [2026/04/23]. https://arxiv.org/abs/2402.14740. arXiv:2402.14740.

275 **LEE H, PHATALE S, MANSOOR H, et al.** RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback[PP/OL]. arXiv. (2023/09/01) [2026/04/23]. https://arxiv.org/abs/2309.00267. arXiv:2309.00267.

276 **AZAR M G, ROWLAND M, PIOT B, et al.** A General Theoretical Paradigm to Understand Learning from Human Preferences[PP/OL]. arXiv. (2023/10/18) [2026/04/23]. https://arxiv.org/abs/2310.12036. arXiv:2310.12036.

277 **YUAN W, PANG R Y, CHO K, et al.** Self-Rewarding Language Models[PP/OL]. arXiv. (2024/01/18) [2026/04/23]. https://arxiv.org/abs/2401.10020. arXiv:2401.10020.

278 **GUO S, ZHANG B, LIU T, et al.** Direct Language Model Alignment from Online AI Feedback [PP/OL]. arXiv. (2024/02/07) [2026/04/23]. https://arxiv.org/abs/2402.04792. arXiv:2402.04792.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

279 **XU S, FU W, GAO J, et al.** Is DPO Superior to PPO for LLM Alignment? A Comprehensive Study [PP/OL]. arXiv. (2024/04/16) [2026/04/23]. https://arxiv.org/abs/2404.10719. arXiv:2404.10719.

280 **ETHAYARAJH K, XU W, MUENNIGHOFF N, et al.** KTO: Model Alignment as Prospect Theoretic Optimization[PP/OL]. arXiv. (2024/02/02) [2026/04/23]. https://arxiv.org/abs/2402.01306. arXiv:2402.01306.

281 **FENG D, QIN B, HUANG C, et al.** Towards Analyzing and Understanding the Limitations of DPO: A Theoretical Perspective[PP/OL]. arXiv. (2024/04/06) [2026/04/23]. https://arxiv.org/abs/2404.04626. arXiv:2404.04626.

282 **PAL A, KARKHANIS D, DOOLEY S, et al.** Smaug: Fixing Failure Modes of Preference Optimisation with DPO-Positive[PP/OL]. arXiv. (2024/02/20) [2026/04/23]. https://arxiv.org/abs/2402.13228. arXiv:2402.13228.

283 **MENG Y, XIA M, CHEN D.** SimPO: Simple Preference Optimization with a Reference-Free Reward [PP/OL]. arXiv. (2024/05/23) [2026/04/23]. https://arxiv.org/abs/2405.14734. arXiv:2405.14734.

284 **XIAO W, WANG Z, GAN L, et al.** A Comprehensive Survey of Direct Preference Optimization: Datasets, Theories, Variants, and Applications[PP/OL]. arXiv. (2024/10/21) [2026/04/23]. https://arxiv.org/abs/2410.15595. arXiv:2410.15595.

285 **YUAN Z, YUAN H, TAN C, et al.** RRHF: Rank Responses to Align Language Models with Human Feedback without tears[PP/OL]. arXiv. (2023/04/11) [2026/04/23]. https://arxiv.org/abs/2304.05302. arXiv:2304.05302.

286 **ZHAO Y, JOSHI R, LIU T, et al.** SLiC-HF: Sequence Likelihood Calibration with Human Feedback [PP/OL]. arXiv. (2023/05/17) [2026/04/23]. https://arxiv.org/abs/2305.10425. arXiv:2305.10425.

287 **XU H, SHARAF A, CHEN Y, et al.** Contrastive Preference Optimization: Pushing the Boundaries of LLM Performance in Machine Translation[PP/OL]. arXiv. (2024/01/16) [2026/04/23]. https://arxiv.org/abs/2401.08417. arXiv:2401.08417.

288 **HONG J, LEE N, THORNE J.** ORPO: Monolithic Preference Optimization without Reference Model[PP/OL]. arXiv. (2024/03/12) [2026/04/23]. https://arxiv.org/abs/2403.07691. arXiv:2403.07691.

289 **PARK R, RAFAILOV R, ERMON S, FINN C.** Disentangling Length from Quality in Direct Preference Optimization[PP/OL]. arXiv. (2024/03/28) [2026/04/23]. https://arxiv.org/abs/2403.19159. arXiv:2403.19159.

290 **ZELIKMAN E, WU Y, MU J, GOODMAN N D.** STaR: Bootstrapping Reasoning With Reasoning [PP/OL]. arXiv. (2022/03/28) [2026/04/23]. https://arxiv.org/abs/2203.14465. arXiv:2203.14465.

291 **UESATO J, KUSHMAN N, KUMAR R, et al.** Solving math word problems with process- and outcome-based feedback[PP/OL]. arXiv. (2022/11/25) [2026/04/23]. https://arxiv.org/abs/2211.14275. arXiv:2211.14275.

292 **HAO S, GU Y, MA H, et al.** Reasoning with Language Model is Planning with World Model[PP/OL]. arXiv. (2023/05/24) [2026/04/23]. https://arxiv.org/abs/2305.14992. arXiv:2305.14992.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

293 **QI Z, MA M, XU J, et al.** Mutual Reasoning Makes Smaller LLMs Stronger Problem-Solvers [PP/OL]. arXiv. (2024/08/12) [2026/04/23]. https://arxiv.org/abs/2408.06195. arXiv:2408.06195.

294 **SCHULMAN J.** Approximating KL Divergence[Z/OL]. (2020/03/07) [2026/04/22]. http://joschu.net/blog/kl-approx.html.

295 **LIANG X, SONG S, ZHENG Z, et al.** Internal Consistency and Self-Feedback in Large Language Models: A Survey[PP/OL]. arXiv. (2024/07/19) [2026/04/23]. https://arxiv.org/abs/2407.14507. arXiv:2407.14507.

296 **YUE Y, CHEN Z, LU R, et al.** Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?[PP/OL]. arXiv. (2025/04/18) [2026/04/23]. https://arxiv.org/abs/2504.13837. arXiv:2504.13837.

297 **XIE T, GAO Z, REN Q, et al.** Logic-RL: Unleashing LLM Reasoning with Rule-Based Reinforcement Learning[PP/OL]. arXiv. (2025/02/20) [2026/04/23]. https://arxiv.org/abs/2502.14768. arXiv:2502.14768.

298 **HUAN M, LI Y, ZHENG T, et al.** Does Math Reasoning Improve General LLM Capabilities? Understanding Transferability of LLM Reasoning[PP/OL]. arXiv. (2025/07/01) [2026/04/23]. https://arxiv.org/abs/2507.00432. arXiv:2507.00432.

299 **YAO Z, LIU Y, CHEN Y, et al.** Are Reasoning Models More Prone to Hallucination?[PP/OL]. arXiv. (2025/05/29) [2026/04/23]. https://arxiv.org/abs/2505.23646. arXiv:2505.23646.

300 **XU F, HAO Q, ZONG Z, et al.** Towards Large Reasoning Models: A Survey of Reinforced Reasoning with Large Language Models[PP/OL]. arXiv. (2025/01/16) [2026/04/23]. https://arxiv.org/abs/2501.09686. arXiv:2501.09686.

301 **DUAN J, ZHANG S, WANG Z, et al.** Efficient Training of Large Language Models on Distributed Infrastructures: A Survey[PP/OL]. arXiv. (2024/07/29) [2026/04/23]. https://arxiv.org/abs/2407.20018. arXiv:2407.20018.

302 **LI S, ZHAO Y, VARMA R, et al.** PyTorch Distributed: Experiences on Accelerating Data Parallel Training[PP/OL]. arXiv. (2020/06/28) [2026/04/23]. https://arxiv.org/abs/2006.15704. arXiv:2006.15704.

303 **SHOEYBI M, PATWARY M, PURI R, et al.** Megatron-LM: Training Multi-Billion Parameter Language Models Using Model Parallelism[PP/OL]. arXiv. (2019/09/17) [2026/04/23]. https://arxiv.org/abs/1909.08053. arXiv:1909.08053.

304 **HUANG Y, CHENG Y, BAPNA A, et al.** GPipe: Efficient Training of Giant Neural Networks using Pipeline Parallelism[PP/OL]. arXiv. (2018/11/16) [2026/04/23]. https://arxiv.org/abs/1811.06965. arXiv:1811.06965.

305 **HARLAP A, NARAYANAN D, PHANISHAYEE A, et al.** PipeDream: Fast and Efficient Pipeline Parallel DNN Training[PP/OL]. arXiv. (2018/06/08) [2026/04/23]. https://arxiv.org/abs/1806.03377. arXiv:1806.03377.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

306 **LIU H, ZAHARIA M, ABBEEL P.** Ring Attention with Blockwise Transformers for Near-Infinite Context[PP/OL]. arXiv. (2023/10/03) [2026/04/23]. https://arxiv.org/abs/2310.01889. arXiv:2310.01889.

307 **LEPIKHIN D, LEE H, XU Y, et al.** GShard: Scaling Giant Models with Conditional Computation and Automatic Sharding[PP/OL]. arXiv. (2020/06/30) [2026/04/23]. https://arxiv.org/abs/2006.16668. arXiv:2006.16668.

308 **XIAO Y, ZHOU Z, MAO F, et al.** An Adaptive Placement and Parallelism Framework for Accelerating RLHF Training[PP/OL]. arXiv. (2023/12/19) [2026/04/23]. https://arxiv.org/abs/2312.11819. arXiv:2312.11819.

309 **CHEN T, XU B, ZHANG C, GUESTRIN C.** Training Deep Nets with Sublinear Memory Cost [PP/OL]. arXiv. (2016/04/21) [2026/04/23]. https://arxiv.org/abs/1604.06174. arXiv:1604.06174.

310 **MICIKEVICIUS P, NARANG S, ALBEN J, et al.** Mixed Precision Training[PP/OL]. arXiv. (2017/10/10) [2026/04/23]. https://arxiv.org/abs/1710.03740. arXiv:1710.03740.

311 **PENG H, WU K, WEI Y, et al.** FP8-LM: Training FP8 Large Language Models[PP/OL]. arXiv. (2023/10/27) [2026/04/23]. https://arxiv.org/abs/2310.18313. arXiv:2310.18313.

312 **HSU P L, DAI Y, KOTHAPALLI V, et al.** Liger Kernel: Efficient Triton Kernels for LLM Training [PP/OL]. arXiv. (2024/10/14) [2026/04/23]. https://arxiv.org/abs/2410.10989. arXiv:2410.10989.

313 **MORITZ P, NISHIHARA R, WANG S, et al.** Ray: A distributed framework for emerging AI applications[C]//13th USENIX Symposium on Operating Systems Design and Implementation. San Diego: USENIX Association, 2018: 561-577.

314 **QI P, WAN X, HUANG G, LIN M.** Zero Bubble Pipeline Parallelism[PP/OL]. arXiv. (2023/11/30) [2026/04/23]. https://arxiv.org/abs/2401.10241. arXiv:2401.10241.

315 **WIJMANS E, HUVAL B, HERTZBERG A, et al.** Cut Your Losses in Large-Vocabulary Language Models[PP/OL]. arXiv. (2024/11/13) [2026/04/23]. https://arxiv.org/abs/2411.09009. arXiv:2411.09009.

316 **LI S, HOEFLER T.** Chimera: Efficiently Training Large-Scale Neural Networks with Bidirectional Pipelines[PP/OL]. arXiv. (2021/07/14) [2026/04/23]. https://arxiv.org/abs/2107.06925. arXiv:2107.06925.

317 **JANG I.** Introducing Context Parallelism[Z/OL]. (2024/09/20) [2025/05/19]. https://insujang.github.io/2024-09-20/introducing-context-parallelism/.

318 **LI D, SHAO R, XIE A, et al.** DISTFLASHATTN: Distributed Memory-efficient Attention for Long-context LLMs Training[PP/OL]. arXiv. (2023/10/05) [2026/04/23]. https://arxiv.org/abs/2310.03294. arXiv:2310.03294.

319 **BRANDON W, NRUSIMHA A, QIAN K, et al.** Striped Attention: Faster Ring Attention for Causal Transformers[PP/OL]. arXiv. (2023/11/15) [2026/04/23]. https://arxiv.org/abs/2311.09431. arXiv:2311.09431.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

320 **REN J, RAJBHANDARI S, AMINABADI R Y, et al.** ZeRO-Offload: Democratizing Billion-Scale Model Training[PP/OL]. arXiv. (2021/01/18) [2026/04/23]. https://arxiv.org/abs/2101.06840. arXiv:2101.06840.

321 **RAJBHANDARI S, RUWASE O, RASLEY J, et al.** ZeRO-Infinity: Breaking the GPU Memory Wall for Extreme Scale Deep Learning[PP/OL]. arXiv. (2021/04/16) [2026/04/23]. https://arxiv.org/abs/2104.07857. arXiv:2104.07857.

322 **SHENG G, ZHANG C, YE Z, et al.** HybridFlow: A Flexible and Efficient RLHF Framework[PP/OL]. arXiv. (2024/09/28) [2026/04/23]. https://arxiv.org/abs/2409.19256. arXiv:2409.19256.

323 **FU W, GAO J, SHEN X, et al.** AReaL: A Large-Scale Asynchronous Reinforcement Learning System for Language Reasoning[PP/OL]. arXiv. (2025/05/30) [2026/04/23]. https://arxiv.org/abs/2505.24298. arXiv:2505.24298.

324 **YU G I, JEONG J S, KIM G W, et al.** Orca: A Distributed Serving System for Transformer-Based Generative Models[C]//16th USENIX Symposium on Operating Systems Design and Implementa-tion. Carlsbad: USENIX Association, 2022: 521-538.

325 **XIAO G, LIN J, SEZNEC M, et al.** SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models[PP/OL]. arXiv. (2022/11/18) [2026/04/23]. https://arxiv.org/abs/2211.10438. arXiv:2211.10438.

326 **LIN J, TANG J, TANG H, et al.** AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration[PP/OL]. arXiv. (2023/06/01) [2026/04/23]. https://arxiv.org/abs/2306.00978. arXiv:2306.00978.

327 **ZHENG L, YIN L, XIE Z, et al.** SGLang: Efficient Execution of Structured Language Model Programs[PP/OL]. arXiv. (2023/12/12) [2026/04/23]. https://arxiv.org/abs/2312.07104. arXiv:2312.07104.

328 **QIN R, LI Z, HE W, et al.** Mooncake: A KVCache-centric Disaggregated Architecture for LLM Serving[PP/OL]. arXiv. (2024/06/24) [2026/04/23]. https://arxiv.org/abs/2407.00079. arXiv:2407.00079.

329 **AGRAWAL A, PANWAR A, MOHAN J, et al.** SARATHI: Efficient LLM Inference by Piggybacking Decodes with Chunked Prefills[PP/OL]. arXiv. (2023/08/31) [2026/04/23]. https://arxiv.org/abs/2308.16369. arXiv:2308.16369.

330 **XIONG W, OGUZ B, GUPTA A, et al.** Simple Local Attentions Remain Competitive for Long-Context Tasks[PP/OL]. arXiv. (2021/12/14) [2026/04/23]. https://arxiv.org/abs/2112.07210. arXiv:2112.07210.

331 **RABE M N, STAATS C.** Self-attention Does Not Need O(n2) Memory[PP/OL]. arXiv. (2021/12/10) [2026/04/23]. https://arxiv.org/abs/2112.05682. arXiv:2112.05682.

332 **MILAKOV M, GIMELSHEIN N.** Online normalizer calculation for softmax[PP/OL]. arXiv. (2018/05/08) [2026/04/23]. https://arxiv.org/abs/1805.02867. arXiv:1805.02867.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致

参考文献

333 **FRANTAR E, ASHKBOOS S, HOEFLER T, ALISTARH D.** GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers[PP/OL]. arXiv. (2022/10/31) [2026/04/23]. https://arxiv.org/abs/2210.17323. arXiv:2210.17323.

334 **WU X, LI C, AMINABADI R Y, et al.** Understanding INT4 Quantization for Transformer Models: Latency Speedup, Composability, and Failure Cases[PP/OL]. arXiv. (2023/01/27) [2026/04/23]. https://arxiv.org/abs/2301.12017. arXiv:2301.12017.

335 **CAI T, LI Y, GENG Z, et al.** Medusa: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads[PP/OL]. arXiv. (2024/01/19) [2026/04/23]. https://arxiv.org/abs/2401.10774. arXiv:2401.10774.

336 **LI Y, WEI F, ZHANG C, ZHANG H.** EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty[PP/OL]. arXiv. (2024/01/26) [2026/04/23]. https://arxiv.org/abs/2401.15077. arXiv:2401.15077.

337 **LI Y, WEI F, ZHANG C, ZHANG H.** EAGLE-2: Faster Inference of Language Models with Dynamic Draft Trees[PP/OL]. arXiv. (2024/06/24) [2026/04/23]. https://arxiv.org/abs/2406.16858. arXiv:2406.16858.

依据 《大语言模型：训练、精调与对齐》书末参考文献表，编号与正文 [n] 一致